1 Chapter 1 Exercises

Not available.

2 Chapter 2 Exercises

Consider the least squares loss expression

$$L[\phi] = \sum_{i \in I} (f[x_i, \phi] - y_i)^2$$

for the linear regression of two variables,

$$f[x_i, \phi] = \begin{bmatrix} 1 & x_i \end{bmatrix} \begin{bmatrix} \phi_0 \\ \phi_1 \end{bmatrix} = X\phi$$

Exercise 2.1. Calculate the partial derivative of f with respect to each ϕ_i, known as the gradient.

Observe the gradient of f is

$$\begin{aligned}
\nabla f[x_i, \phi] &= \left\langle \frac{\partial f}{\partial \phi_0}, \frac{\partial f}{\partial \phi_1} \right\rangle \\
&= \left\langle \begin{bmatrix} 1 & x_i \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & x_i \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\rangle \\
&= \langle 1, x_i \rangle
\end{aligned}$$

Exercise 2.2. Find the minima of $L[\phi]$.

The minima occur at the critical points of L, namely where $\nabla_\phi L = 0$. Since nabla is a linear operator, we can pass it over the sums and also invoke the chain rule, i.e.,

$$\begin{aligned}
\nabla_\phi L[\phi] &= \nabla_\phi \left(\sum_{i \in I} (f[x_i, \phi] - y_i)^2 \right) \\
&= \sum_{i \in I} \nabla_\phi (f[x_i, \phi] - y_i)^2 \\
&= \sum_{i \in I} (2(f[x_i, \phi] - y_i) \cdot \nabla_\phi (f[x_i, \phi] - y_i)) \\
&= \sum_{i \in I} (2(f[x_i, \phi] - y_i) \cdot (\nabla_\phi f[x_i, \phi] - \nabla y_i)) \\
&= \sum_{i \in I} (2(f[x_i, \phi] - y_i) \cdot \langle 1, x_i \rangle)
\end{aligned}$$

Thus

$$\begin{aligned}
\nabla_\phi L[\phi] &= \sum_{i \in I} (2(f[x_i, \phi] - y_i) \cdot \langle 1, x_i \rangle) \\
&= \sum_{i \in I} \langle 2(f[x_i, \phi] - y_i), 2(f[x_i, \phi] - y_i) x_i \rangle \\
&= 2 \sum_{i \in I} \langle f[x_i, \phi] - y_i, (f[x_i, \phi] - y_i) x_i \rangle
\end{aligned}$$

Next, to find the critical points, $\nabla_\phi L[\phi] = \widehat{0}$ occurs for each i,

$$2\sum_{i\in I}\left\langle f\left[x_i,\phi\right]-y_i,\left(f\left[x_i,\phi\right]-y_i\right)x_i\right\rangle=\langle 0,0\rangle$$

for the first entry, we have

$$\begin{aligned}\sum_{i\in I}\left(f\left[x_i,\phi\right]-y_i\right)&=0\\ \sum_{i\in I}\left(\left[\begin{array}{cc}1 & x_i\end{array}\right]\left[\begin{array}{c}\phi_0\\ \phi_1\end{array}\right]-y_i\right)&=0\\ \sum_{i\in I}\left(\left[\begin{array}{cc}1 & x_i\end{array}\right]\left[\begin{array}{c}\phi_0\\ \phi_1\end{array}\right]\right)-\sum_{i\in I}y_i&=0\end{aligned}$$

and solving for ϕ_0 yields

$$\begin{aligned}\sum_{i\in I}\left(\phi_0+x_i\phi_1\right)-\sum_{i\in I}y_i&=0\\ \sum_{i\in I}\phi_0+\sum_{i\in I}x_i\phi_1-\sum_{i\in I}y_i&=0\\ \phi_0\left|I\right|+\phi_1\sum_{i\in I}x_i-\sum_{i\in I}y_i&=0\\ \frac{\sum_{i\in I}y_i-\phi_1\sum_{i\in I}x_i}{\left|I\right|}&=\phi_0\\ \overline{y}-\phi_1\overline{x}&=\phi_0\end{aligned}$$

For the second entry, we have

$$\begin{aligned}\sum_{i\in I}\left(f\left[x_i,\phi\right]-y_i\right)x_i&=0\\ \sum_{i\in I}\left(\left[\begin{array}{cc}1 & x_i\end{array}\right]\left[\begin{array}{c}\phi_0\\ \phi_1\end{array}\right]-y_i\right)x_i&=0\\ \sum_{i\in I}\left(\left[\begin{array}{cc}x_i & x_i^2\end{array}\right]\left[\begin{array}{c}\phi_0\\ \phi_1\end{array}\right]-x_iy_i\right)&=0\end{aligned}$$

and solving for $\phi_1$yields

$$\begin{aligned}\sum_{i\in I}\left(x_i\phi_0+x_i^2\phi_1-x_iy_i\right)&=0\\ \sum_{i\in I}x_i\phi_0+\sum_{i\in I}x_i^2\phi_1-\sum_{i\in I}x_iy_i&=0\\ \phi_0\sum_{i\in I}x_i+\phi_1\sum_{i\in I}x_i^2-\sum_{i\in I}x_iy_i&=0\end{aligned}$$

substituting ϕ_0 yields with

$$\begin{aligned}\left(\overline{y}-\phi_1\overline{x}\right)\sum_{i\in I}x_i+\phi_1\sum_{i\in I}x_i^2-\sum_{i\in I}x_iy_i&=0\\ \overline{y}\sum_{i\in I}x_i-\phi_1\overline{x}\sum_{i\in I}x_i+\phi_1\sum_{i\in I}x_i^2-\sum_{i\in I}x_iy_i&=0\\ \frac{\overline{y}\sum_{i\in I}x_i-\sum_{i\in I}x_iy_i}{\left(\overline{x}\sum_{i\in I}x_i-\sum_{i\in I}x_i^2\right)}&=\phi_1\end{aligned}$$

Rewrite the numerator as

$$\begin{aligned}
\overline{y}\sum_{i\in I} x_i - \sum_{i\in I} x_i y_i &= \overline{y}\sum_{i\in I} x_i - \sum_{i\in I} x_i y_i + 0 \\
&= \overline{y}\sum_{i\in I} x_i - \sum_{i\in I} x_i y_i + \left(\overline{y}\sum_{i\in I} x_i - \overline{y}\sum_{i\in I} x_i\right) \\
&= \overline{y}\sum_{i\in I} x_i - \sum_{i\in I} x_i y_i + \frac{\sum_{i\in I} y_i}{|I|}\sum_{i\in I} x_i - |I|\,\overline{y}\frac{\sum_{i\in I} x_i}{|I|} \\
&= \overline{y}\sum_{i\in I} x_i - \sum_{i\in I} x_i y_i + \overline{x}\sum_{i\in I} y_i - |I|\,\overline{yx} \\
&= \sum_{i\in I} \overline{y}x_i - \sum_{i\in I} x_i y_i + \sum_{i\in I} \overline{x}y_i - \sum_{i\in I} \overline{yx} \\
&= \sum_{i\in I} \left(\overline{y}x_i - x_i y_i + \overline{x}y_i - \overline{yx}\right) \\
&= \sum_{i\in I} \left(x_i\left(\overline{y} - y_i\right) - \overline{x}\left(\overline{y} - y_i\right)\right) \\
&= -\sum_{i\in I} \left(x_i - \overline{x}\right)\left(y_i - \overline{y}\right)
\end{aligned}$$

Rewrite the denominator as

$$\begin{aligned}
\overline{x}\sum_{i\in I} x_i - \sum_{i\in I} x_i^2 &= 2\overline{x}\sum_{i\in I} x_i - \overline{x}\sum_{i\in I} x_i - \sum_{i\in I} x_i^2 \\
&= \sum_{i\in I} 2\overline{x}x_i - \sum_{i\in I} \overline{x}^2 - \sum_{i\in I} x_i^2 \\
&= -\sum_{i\in I} \left(x_i^2 - 2\overline{x}x_i + \overline{x}^2\right) \\
&= -\sum_{i\in I} \left(x_i - \overline{x}\right)^2
\end{aligned}$$

Finally,

$$\phi_1 = \frac{\overline{y}\sum_{i\in I} x_i - \sum_{i\in I} x_i y_i}{\overline{x}\sum_{i\in I} x_i - \sum_{i\in I} x_i^2} = \frac{\sum_{i\in I}\left(x_i - \overline{x}\right)\left(y_i - \overline{y}\right)}{\sum_{i\in I}\left(x_i - \overline{x}\right)^2}$$

Hence,

$$\widehat{\phi} = \begin{bmatrix} \overline{y} - \phi_1\overline{x} \\ \frac{\sum_{i\in I}(x_i - \overline{x})(y_i - \overline{y})}{\sum_{i\in I}(x_i - \overline{x})^2} \end{bmatrix}$$

Exercise 2.3. Reformulate the linear regression as a generative model, i.e.,

$$x = g\left[y, \phi\right] = \begin{bmatrix} 1 & y \end{bmatrix}\begin{bmatrix} \phi_0 \\ \phi_1 \end{bmatrix}$$

(a) Find the new loss function and the inference expression, $y = g^{-1}\left[x, \phi\right]$.

(b) Verify if the model will make the same predictions as the discriminate version. Use code if you need to.

We'll solve in parts.

(a) Since least squares loss is defined using the difference between the input and the target variables, the least squares loss function for the generative model $x = g\left[y, \phi\right]$ can be expressed as

$$L\left[\phi\right] = \sum_{i\in I}\left(x_i - g\left[y_i, \phi\right]\right)^2$$

The inference expression can be solved for in terms of y

$$X = Y\phi \implies Y = X\phi^T \left(\phi\phi^T\right)^{-1}$$

But we're lazy, so we'll just expand the inner product and solve for y,

$$\begin{aligned} x &= \left[\begin{array}{cc} 1 & y \end{array}\right] \left[\begin{array}{c} \phi_0 \\ \phi_1 \end{array}\right] = \phi_0 + y\phi_1 \\ \implies y &= \frac{x - \phi_0}{\phi_1} \\ &= \frac{x}{\phi_1} - \frac{\phi_0}{\phi_1} \\ &= \left[\begin{array}{cc} 1 & x \end{array}\right] \left[\begin{array}{c} -\frac{\phi_0}{\phi_1} \\ \frac{1}{\phi_1} \end{array}\right] \end{aligned}$$

where $\phi_1 \neq 0$

(b) Fitting $y \sim f[\phi, x]$ via linear regression wouldn't be the same as fitting $x \sim g[\theta, x]$ unless their parameters satisfy

$$\widehat{\phi}_1 = \frac{1}{\widehat{\theta}_1}$$

or, specifically,

$$\frac{Cov\,(x, y)}{Var\,(y)} = \frac{1}{\frac{Cov(x,y)}{Var(x)}}$$

This happens when the modeled data is strictly linear.

3 Chapter 3 Exercises

Definition 3.1. A shallow neural network is a neural network that has exactly one hidden layer.

Consider the following shallow neural network $y = f[x, \{\phi, \theta\}]$, with parameters ϕ_{ij}, θ_{ij}, inputs x, and activation function a,

$$\begin{aligned} y &= \phi_0 + \phi_1 a[\theta_{10} + \theta_{11}x] + \phi_2 a[\theta_{20} + \theta_{21}x] + \phi_3 a[\theta_{30} + \theta_{31}x] \\ &= \phi_0 + \sum_{i=1}^{3} \phi_i a[\theta_{i0} + \theta_{i1}x] \end{aligned}$$

This function has one input, one output, and three hidden units.

Exercise 3.1. What would y look like if a was

(a) linear, i.e., $a[z] = \psi_0 + \psi_1 z$?

(b) constant, i.e., $a[z] = z$?

We'll solve case-by-case.

(a) If a is linear, then

$$\begin{aligned} y &= \phi_0 + \sum_{i=1}^{3} \phi_i \left(\psi_0 + \psi_1\left(\theta_{i0} + \theta_{i1}x\right)\right) \\ &= \phi_0 + \sum_{i=1}^{3} \phi_i\psi_0 + \sum_{i=1}^{3} \phi_i\psi_1\left(\theta_{i0} + \theta_{i1}x\right) \\ &= \phi_0 + \psi_0 \sum_{i=1}^{3} \phi_i + \psi_1 \sum_{i=1}^{3} \phi_i\left(\theta_{i0} + \theta_{i1}x_i\right) \end{aligned}$$

(b) If a is constant, then

$$y = \phi_0 + \sum_{i=1}^{3} \phi_i \left(\theta_{i0} + \theta_{i1} x\right)$$

Consider Figure 3.3, which shows the preactivations (inputs), activations (outputs) on ReLU, and the weighed sum.

Exercise 3.2. Determine which activations contributed (active) or did not contribute (inactive) as a result of a clipped ReLU output in Figure 3.3.

The "joints" in the j) highlight the specific cutoff points for each region. Let's mark them appropriately.

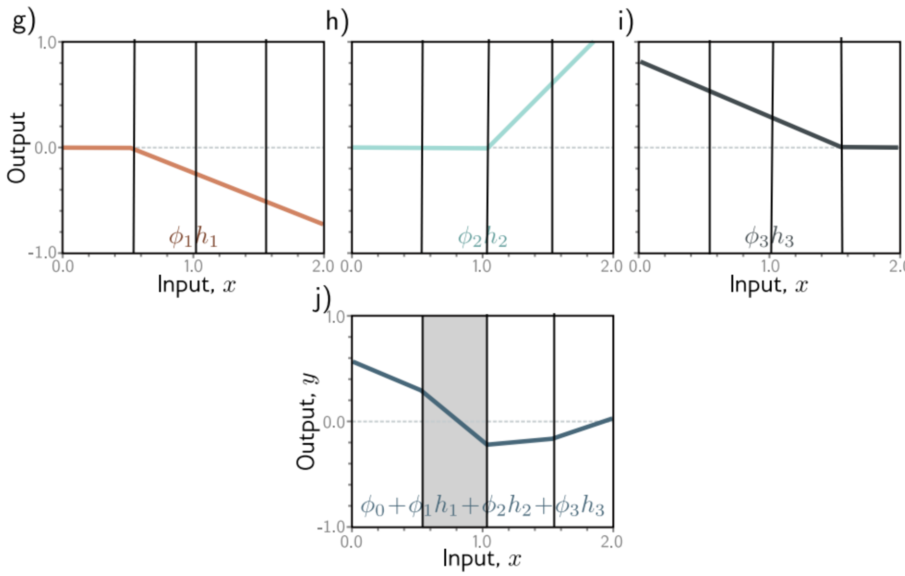

Figure 1: Modified version of figure 3.3 with the regions marked based on the j) joints. All rights reserved to the original author.

From here, we can see when numbering the regions from left to right,

- region1: g) is clipped (inactive), h) is clipped (inactive), i) is not clipped (active)
- region2: g) is not clipped (active), h) is clipped (inactive), i) is not clipped (active)
- region3: g) is not clipped (active), h) is not clipped (active), i) is not clipped (active)
- region4: g) is not clipped (active), h) is not clipped (active), i) is clipped (inactive)

Exercise 3.3. Derive expressions for the positions of the "joints" in function in figure 3.3j in terms of $\{\phi_{ij}, \theta_{ij}\}$ and x. Derive expressions for the slopes of the four linear regions.

A "joint" occurs when clipping occurs, i.e., when the activation goes to zero. For ReLU, this occurs when the inputs are negative.

Observe each θx input is in the set

$$I = \{\theta_{i0} + \theta_{i1} x : i = 1, \ldots, n\}$$

So setting $I \leq 0$ and solving for x yields the inactive regions,

$$\begin{aligned} & I \leq 0 \\ \implies & \{\theta_{i0} + \theta_{i1} x \leq 0 : i = 1, \ldots, n\} \\ \implies & \left\{ x \leq -\frac{\theta_{i0}}{\theta_{i1}} : i = 1, \ldots, n; \theta_{i1} \neq 0 \right\} \end{aligned}$$

The set $I = 0$ defines the critical points in which the joints occur.

Then calculating $y = f\left[\{\phi_{ij}, \theta_{ij}\}, I = 0\right]$ yields the points

$$y(I=0) = \begin{bmatrix} y(x_1) \\ y(x_2) \\ y(x_3) \end{bmatrix} = \begin{bmatrix} \phi_0 + \sum_{i=1}^{3} \phi_i a(\theta_{i0} + \theta_{i1} x_1) \\ \phi_0 + \sum_{i=1}^{3} \phi_i a(\theta_{i0} + \theta_{i1} x_2) \\ \phi_0 + \sum_{i=1}^{3} \phi_i a(\theta_{i0} + \theta_{i1} x_3) \end{bmatrix}$$
$$= \begin{bmatrix} \phi_0 + \phi_1 a(0) + \sum_{i\in\{2,3\}} \phi_i a(\theta_{i0} + \theta_{i1} x_1) \\ \phi_0 + \phi_2 a(0) + \sum_{i\in\{1,3\}} \phi_i a(\theta_{i0} + \theta_{i1} x_2) \\ \phi_0 + \phi_3 a(0) + \sum_{i\in\{1,2\}} \phi_i a(\theta_{i0} + \theta_{i1} x_3) \end{bmatrix}$$
$$= \begin{bmatrix} \phi_0 + \sum_{i\in\{2,3\}} \phi_i a(\theta_{i0} + \theta_{i1} x_1) \\ \phi_0 + \sum_{i\in\{1,3\}} \phi_i a(\theta_{i0} + \theta_{i1} x_2) \\ \phi_0 + \sum_{i\in\{1,2\}} \phi_i a(\theta_{i0} + \theta_{i1} x_3) \end{bmatrix}$$

To derive the slopes for the four linear regions, we need to calculate the general slope expression.

The derivative of the ReLU activation function $a(\cdot)$ is

$$\frac{d}{dx} ReLU(x) = \frac{d}{dx} \max(0, x)$$

Let's approach this using the definition of the derivative around 0. Observe

$$\lim_{h\to 0^-} \frac{\max(0,h)}{h} = \lim_{h\to 0^-} \frac{0}{h} = 0$$

$$\lim_{h\to 0^+} \frac{\max(0,h)}{h} = \lim_{h\to 0^+} \frac{h}{h} = 1$$

So

$$\frac{d}{dx}\max(0,x) = \begin{cases} 0 & x < 0 \\ DNE & x = 0 \\ 1 & 0 < x \end{cases}$$

Then since derivatives are linear and the chain rule implies

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx}\left(\phi_0 + \sum_{i=1}^{3} \phi_i a(\theta_{i0} + \theta_{i1} x)\right) \\ &= \frac{d}{dx}\phi_0 + \sum_{i=1}^{3} \phi_i \frac{d}{dx} a(\cdot) \frac{d}{dx}(\theta_{i0} + \theta_{i1} x) \\ &= \frac{d}{dx}\phi_0 + \sum_{i=1}^{3} \phi_i \theta_{i1} \frac{d}{dx} a(\cdot) \\ &= \sum_{i=1}^{3} \phi_i \theta_{i1} 1_{0<(\theta_{i0}+\theta_{i1}x)} \end{aligned}$$

where 1 is the indicator function.

Finally, since there are three critical points (three joints) in $I = 0$, then we have four regions. We can calculate each region's slope based on our answers in Exercise 2, since clipped values will take $1_{0<(\theta_{i0}+\theta_{i1}x)} = 0$.

- region1: g) is clipped (inactive), h) is clipped (inactive), i) is not clipped (active) implies

$$\sum_{i=1}^{3} \phi_i \theta_{i1} 1_{0<(\theta_{i0}+\theta_{i1}x)} = \phi_3 \theta_{31}$$

- region2: g) is not clipped (active), h) is clipped (inactive), i) is not clipped (active) implies

$$\sum_{i=1}^{3} \phi_i \theta_{i1} 1_{0<(\theta_{i0}+\theta_{i1}x)} = \phi_1\theta_{11} + \phi_3\theta_{31}$$

- region3: g) is not clipped (active), h) is not clipped (active), i) is not clipped (active) implies

$$\sum_{i=1}^{3} \phi_i \theta_{i1} 1_{0<(\theta_{i0}+\theta_{i1}x)} = \sum_{i=1}^{3} \phi_i\theta_{i1}$$

- region4: g) is not clipped (active), h) is not clipped (active), i) is clipped (inactive) implies

$$\sum_{i=1}^{3} \phi_i \theta_{i1} 1_{0<(\theta_{i0}+\theta_{i1}x)} = \sum_{i=1}^{2} \phi_i\theta_{i1}$$

Exercise 3.4. Skip

Exercise 3.5. Prove for scalar $\alpha \in \mathbb{R}^+$, $ReLU(\alpha x) = \alpha ReLU(x)$.

Observe α is positive implies scaling inequalities are preserved, thus

$$ReLU(\alpha x) = \max(0, \alpha x) = \begin{cases} \alpha x & 0 < \alpha x \\ 0 & \alpha x \le 0 \end{cases} = \alpha \begin{cases} x & 0 < x \\ 0 & x \le 0 \end{cases} = \alpha \max(0, x) = \alpha ReLU(x)$$

Exercise 3.6. Let

$$\frac{1}{\alpha}\phi_1 h_1 = \frac{1}{\alpha} a(\alpha(\theta_{10} + \theta_{11}x))$$

for some scalar $\alpha \in \mathbb{R} - \{0\}$ and $a = ReLU$. Explore the expression depending on $sign(\alpha)$.

Observe

$$\begin{aligned} \frac{1}{\alpha}\phi_1 h_1 &= \frac{1}{\alpha} ReLU(\alpha(\theta_{10} + \theta_{11}x)) \\ &= \frac{1}{\alpha} \max(0, \alpha(\theta_{10} + \theta_{11}x)) \\ &= \frac{1}{\alpha} \begin{cases} \alpha(\theta_{10} + \theta_{11}x) & 0 < \alpha(\theta_{10} + \theta_{11}x) \\ 0 & \alpha(\theta_{10} + \theta_{11}x) \le 0 \end{cases} \end{aligned}$$

If $0 < \alpha$

$$\begin{aligned} \frac{1}{\alpha} \begin{cases} \alpha(\theta_{10} + \theta_{11}x) & 0 < \alpha(\theta_{10} + \theta_{11}x) \\ 0 & \alpha(\theta_{10} + \theta_{11}x) \le 0 \end{cases} &= \begin{cases} (\theta_{10} + \theta_{11}x) & 0 < (\theta_{10} + \theta_{11}x) \\ 0 & (\theta_{10} + \theta_{11}x) \le 0 \end{cases} \\ &= ReLU(\theta_{10} + \theta_{11}x) \end{aligned}$$

If $\alpha < 0$,

$$\begin{aligned} \frac{1}{\alpha} \begin{cases} \alpha(\theta_{10} + \theta_{11}x) & 0 < \alpha(\theta_{10} + \theta_{11}x) \\ 0 & \alpha(\theta_{10} + \theta_{11}x) < 0 \end{cases} &= \begin{cases} (\theta_{10} + \theta_{11}x) & (\theta_{10} + \theta_{11}x) < 0 \\ 0 & 0 \le (\theta_{10} + \theta_{11}x) \end{cases} \\ &= \min(0, \theta_{10} + \theta_{11}x) \end{aligned}$$

Hence

$$\frac{1}{\alpha}\phi_1 h_1 = \begin{cases} ReLU(\theta_{10} + \theta_{11}x) & 0 < \alpha \\ \min(0, \theta_{10} + \theta_{11}x) & 0 \le \alpha \end{cases}$$

Exercise 3.7. Consider the least squares function on the shallow network. Does this loss function have a unique minimum? i.e., is there a single "best" set of parameters?

Observe

$$\begin{aligned}
L[\phi,\theta] &= \sum_{j\in I}\left(f\left[x^{(j)},\{\phi,\theta\}\right]-y^{(j)}\right)^2 \\
&= \sum_{j\in I}\left(\phi_0+\sum_{i=1}^{3}\phi_i a\left(\theta_{i0}+\theta_{i1}x^{(j)}\right)-y^{(j)}\right)^2
\end{aligned}$$

We need to differentiate wrt the parameters ϕ_0, and ϕ_i, θ_{i0},θ_{i1} for $i = 1, 2, 3$.

Regardless of the differentiated variable, we will always use the chain rule, i.e.,

$$\begin{aligned}
\frac{\partial L}{\partial(\cdot)} &= \sum_{j\in I}2\left(y-y^{(j)}\right)\frac{\partial}{\partial(\cdot)}\left(\phi_0+\sum_{i=1}^{3}\phi_i a\left(\theta_{i0}+\theta_{i1}x^{(j)}\right)-y^{(j)}\right) \\
&= \sum_{j\in I}2\left(y-y^{(j)}\right)\left(\frac{\partial\phi_0}{\partial(\cdot)}+\frac{\partial}{\partial(\cdot)}\sum_{i=1}^{3}\phi_i a\left(\theta_{i0}+\theta_{i1}x^{(j)}\right)-\frac{\partial y^{(j)}}{\partial(\cdot)}\right) \\
&= \sum_{j\in I}2\left(y-y^{(j)}\right)\left(\frac{\partial\phi_0}{\partial(\cdot)}+\frac{\partial}{\partial(\cdot)}\sum_{i=1}^{3}\phi_i a\left(\theta_{i0}+\theta_{i1}x^{(j)}\right)\right) \\
&= \sum_{j\in I}2\left(y-y^{(j)}\right)\left(\frac{\partial\phi_0}{\partial(\cdot)}+\sum_{i=1}^{3}\frac{\partial}{\partial(\cdot)}\left(\phi_i a\left(\theta_{i0}+\theta_{i1}x^{(j)}\right)\right)\right) \\
&= \sum_{j\in I}2\left(y-y^{(j)}\right)\left(\frac{\partial\phi_0}{\partial(\cdot)}+\sum_{i=1}^{3}\left(\frac{\partial\phi_i}{\partial(\cdot)}a\left(\theta_{i0}+\theta_{i1}x^{(j)}\right)+\phi_i\frac{\partial}{\partial(\cdot)}\left(a\left(\theta_{i0}+\theta_{i1}x^{(j)}\right)\right)\right)\right) \\
&= \sum_{j\in I}2\left(y-y^{(j)}\right)\left(\frac{\partial\phi_0}{\partial(\cdot)}+\sum_{i=1}^{3}\left(\frac{\partial\phi_i}{\partial(\cdot)}a\left(\theta_{i0}+\theta_{i1}x^{(j)}\right)+\phi_i\frac{\partial a\left(\theta_{i0}+\theta_{i1}x^{(j)}\right)}{\partial(\cdot)}\frac{\partial}{\partial(\cdot)}\left(\theta_{i0}+\theta_{i1}x^{(j)}\right)\right)\right) \\
&= \sum_{j\in I}2\left(y-y^{(j)}\right)\left(\frac{\partial\phi_0}{\partial(\cdot)}+\sum_{i=1}^{3}\left(\frac{\partial\phi_i}{\partial(\cdot)}a\left(\theta_{i0}+\theta_{i1}x^{(j)}\right)+\phi_i\frac{\partial a\left(\theta_{i0}+\theta_{i1}x^{(j)}\right)}{\partial(\cdot)}\left(\frac{\partial\theta_{i0}}{\partial(\cdot)}+\frac{\partial\theta_{i1}}{\partial(\cdot)}x^{(j)}\right)\right)\right)
\end{aligned}$$

So it remains to calculate the derivative on the remaining factor in each summmand for each parameter in

$$\frac{\partial\phi_0}{\partial(\cdot)}+\sum_{i=1}^{3}\left(\frac{\partial\phi_i}{\partial(\cdot)}a\left(\theta_{i0}+\theta_{i1}x^{(j)}\right)+\phi_i\frac{\partial a\left(\theta_{i0}+\theta_{i1}x^{(j)}\right)}{\partial(\cdot)}\left(\frac{\partial\theta_{i0}}{\partial(\cdot)}+\frac{\partial\theta_{i1}}{\partial(\cdot)}x^{(j)}\right)\right)$$

For ϕ_0,

$$\frac{\partial\phi_0}{\partial\phi_0}+\frac{\partial}{\partial\phi_0}\sum_{i=1}^{3}\phi_i a\left(\theta_{i0}+\theta_{i1}x^{(j)}\right)=1$$

For ϕ_i,

$$\frac{\partial\phi_0}{\partial\phi_i}+\sum_{i=1}^{3}\left(\frac{\partial\phi_i}{\partial\phi_i}a\left(\theta_{i0}+\theta_{i1}x^{(j)}\right)+\phi_i\frac{\partial a\left(\theta_{i0}+\theta_{i1}x^{(j)}\right)}{\partial\phi_i}\left(\frac{\partial\theta_{i0}}{\partial\phi_i}+\frac{\partial\theta_{i1}}{\partial\phi_i}x^{(j)}\right)\right)=\sum_{i=1}^{3}\left(a\left(\theta_{i0}+\theta_{i1}x^{(j)}\right)\right)$$

For θ_{i0},

$$\frac{\partial\phi_0}{\partial\theta_{i0}}+\sum_{i=1}^{3}\left(\frac{\partial\phi_i}{\partial\theta_{i0}}a\left(\theta_{i0}+\theta_{i1}x^{(j)}\right)+\phi_i\frac{\partial a\left(\theta_{i0}+\theta_{i1}x^{(j)}\right)}{\partial\theta_{i0}}\left(\frac{\partial\theta_{i0}}{\partial\theta_{i0}}+\frac{\partial\theta_{i1}}{\partial\theta_{i0}}x^{(j)}\right)\right)=\sum_{i=1}^{3}\left(\phi_i 1_{\left(\theta_{i0}+\theta_{i1}x^{(j)}\right)}(1)\right)$$

For θ_{i1},

$$\frac{\partial \phi_0}{\partial \theta_{i1}}+\sum_{i=1}^{3}\left(\frac{\partial \phi_i}{\partial \theta_{i1}} a\left(\theta_{i0}+\theta_{i1} x^{(j)}\right)+\phi_i \frac{\partial a\left(\theta_{i0}+\theta_{i1} x^{(j)}\right)}{\partial \theta_{i1}}\left(\frac{\partial \theta_{i0}}{\partial \theta_{i1}}+\frac{\partial \theta_{i1}}{\partial \theta_{i1}} x^{(j)}\right)\right)=\sum_{i=1}^{3}\left(\phi_i 1_{\left(\theta_{i0}+\theta_{i1} x^{(j)}\right)}\left(x^{(j)}\right)\right)$$

The minima really depends on the ReLU's possible non-differentiable behaviour for

$$\left(\theta_{i0}+\theta_{i1} x^{(j)}\right),$$

which introduces the "joints" and implies the space is non-convex. In other words, the shallow network with this activation function has no global minima.

Exercise 3.8. What does the shallow network look like with these activation functions, i.e., heaviside, rectangular, and tanh? Provide an informal description of the family of functions that can be created by neural networks with one input, three hidden units, and one output for each activation function.

Neural networks can model non-linear data with non-linear activation functions because they can bend, stretch, and shift to fit different data types.

Exercise 3.9. Show that the third linear region in Figure 3.3j has a slope that is the sum of the slopes of the first and fourth linear regions.

Is this by chance? Exercise 3.3 shows the final linear regions in Figure 3.3j satisfy

$$\sum_{i=1}^{3} \phi_i \theta_{i1}=\phi_3 \theta_{31}+\sum_{i=1}^{2} \phi_i \theta_{i1}$$

Exercise 3.10. Consider a neural network with one input, one output, and three hidden units. The construction in figure 3.3 shows how this creates four linear regions based on the number of "joints." Under what circumstances could this network produce a function with fewer than four linear regions? Assume the activation function is still $ReLU$.

Recall a "joint" occurs when clipping occurs, i.e., when the activation goes to zero. We used the interactive tool hosted on the website to play around with the inputs and populate the following table to identify some possible reasons for having less than 4 regions.

Number of Regions	Reason
1	All pre-activations are positive (no clipping)
1	All pre-activations are less than or equal to zero (completely clipped)
2	Two pre-activations have a different sign than the third pre-activation
2	One pre-activation has mixed signs, the others are monotonic
2	Two pre-activations are identical (linearly dependent). The other has mixed signs and an opposite slope

Exercise 3.11. Consider a neural network with one input, N outputs, and one layer with H hidden units. How many parameters does it have?

Each hidden unit has 1 weight for each input, and 1 bias, totaling $H \cdot 1+H$. Each output has 1 weight for each hidden unit, so $H \cdot N$ weights, and 1 bias, totaling $H \cdot N+N$.

Thus

$$\begin{aligned}\sum Weights+\sum Biases &=(H+H \cdot N)+(H+N) \\ &=H(2+N)+N\end{aligned}$$

Exercise 3.12. Consider a neural network with I inputs, N outputs, and one layer with H hidden units. How many parameters does it have?

Each hidden unit has 1 weight for each input, and 1 bias, totaling $H \cdot I + H$. Each output has 1 weight for each hidden unit, so $H \cdot N$ weights, and 1 bias, totaling $H \cdot N + N$.

Thus

$$\begin{aligned}\sum Weights + \sum Biases &= (H \cdot I + H \cdot N) + (H + N) \\ &= H\,(I + 1 + N) + N\end{aligned}$$

Exercise 3.13. In figure 3.8, which hidden units are active (pass the input) and which are inactive (clip the input) for each of the seven regions?

Figure 3.8j is the accumulation of the active and inactive regions from Figure 3.8(g-i). Their contributions are summarized in the table.

region	g clipped?	h clipped?	i clipped?
top left	N	Y	Y
top right	N	N	Y
mid left	Y	Y	Y
center	N	Y	N
mid right	N	N	N
bottom left	Y	Y	N
bottom center	Y	N	N

Exercise 3.14. Write out the formula for a neural network with I inputs, one hidden layer of H hidden units, and N outputs.

The formula for the output layer, or vector, can be written as

$$y = \left\{ \phi_{n0} + \sum_{h=1}^{H} \phi_{nh} a \left[\theta_{h0} + \sum_{i=1}^{I} \theta_{hi} x_i \right] \right\}_{j=1}^{N}$$

Exercise 3.15. What is the maximum possible number of linear regions that can be created from a neural network with I inputs, one hidden layer of H hidden units, and N outputs?

Let A be an essential arrangement of hyperplanes in $\mathbb{R}^I$. Recall Zaslavsky's theorem says that the number of regions of A is given by

$$r\,(A) = (-1)^I \chi_A\,(-1)\,,$$

and the number of bounded regions is given by

$$b\,(A) = (-1)^I \chi_A\,(1)$$

in which χ_A is the characteristic polynomial,

$$\chi_A = \sum_{x \in L(A)} \mu\,(0, x)\, t^{\dim(x)}$$

for the intersection lattice $L\,(A)$ and function $\mu\,(0, x)$. If we assume the regions carved out by the neural network are not linearly dependent, and the hyperplanes defined in the intersection lattice follow the inclusion-exclusion principle, in that no hyperplanes intersect at more than I times, then $\mu\,(0, x)$ follows the binomial coefficients, and $\dim\,(x)$ decreases with each intersection, i.e.,

$$\chi_A\,(t) = \sum_{j=0}^{I} (-1)^j \binom{H}{j} t^{I-j}$$

in which

$$\chi_A\,(-1) = (-1)^I \sum_{j=0}^{I} \binom{H}{j}$$

Finally, the number of regions is

$$
\begin{aligned}
r(A) &= (-1)^I \chi_A(-1) \\
&= \sum_{j=0}^{I} \binom{H}{j}
\end{aligned}
$$

Another approach to this is inductively. Let the base case be zero hyperplanes, so there is one region. Observe adding the $(k+1)$th hyperplane can intersect each of the existing k hyperplanes in the $\mathbb{R}^{I-1}$ subspace. If we have $H-1$ hyperplanes and we add one more, then the total number of regions equals the existing regions in plus the new ones created on the new hyperplane in